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HEAT DISSIPATION MODULE AND ANTENNA ARRAY DEVICE HAVING SAME

Abstract

A heat dissipation module and an antenna array device applying the heat dissipation module are provided, the antenna array device includes an upper cover and a circuit board, the heat dissipation module includes a body; a plurality of heat-conducting structures arranged in an array on one side of the body; a plurality of supporting structures supporting the circuit board, the plurality of heat-conducting structures and the plurality of supporting structures arranged on a same side of the body; and a plurality of connecting structures arranged on an edge of the body and connected to the upper cover.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Chinese Patent Application No. 202410274559.7 filed on Mar. 11, 2024, in China National Intellectual Property Administration, the contents of which are incorporated by reference herein.

FIELD

[0002] The subject matter herein generally relates to antenna technology field, and more particularly to a heat dissipation module and an antenna array device having the heat dissipation module.

BACKGROUND

[0003] An antenna system composed of many identical single antennas arranged in a certain pattern is also called an antenna array. A phased array antenna formed by the antenna array can efficiently communicate with low-orbit satellites due to its high flexibility and wide-angle scanning advantages. With the development of low-orbit satellite communication technology, wireless communication devices equipped with antenna arrays that can communicate with low-orbit satellites have also emerged. However, the antenna array has poor heat dissipation performance, especially in the narrow space of the wireless communication device. If the heat generated by the antenna array during operation cannot be quickly dissipated, the signal transmission quality will be affected.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Implementations of the present disclosure will now be described, by way of embodiments, with reference to the attached figures.

[0005] FIG. 1 is a structural diagram of an antenna array device according to an embodiment of the present application.

[0006] FIG. 2 is a partial explosive view of the antenna array device shown in FIG. 1.

[0007] FIG. 3 is a cross-sectional view along an III-III line in FIG. 1.

[0008] FIG. 4 is an explosive view of a heat dissipation module of the antenna array device shown in FIG. 1.

[0009] FIG. 5 is a schematic diagram showing a ratio of an area of several heat-conducting structures to an area of several radiation units.

[0010] FIG. 6 is another structural diagram of the heat dissipation module of the antenna array device shown in FIG. 1.

[0011] FIG. 7 is a cross-sectional view along an VI-VI line in FIG. 1.

[0012] FIG. 8 is a structural diagram of an antenna array device according to another embodiment of the present application.

[0013] FIG. 9 is a structural diagram of heat dissipation module of the antenna array device shown in FIG. 8.

[0014] FIG. 10 is an explosive view of the antenna array device shown in FIG. 8.

[0015] FIG. 11 is a structural diagram of the antenna array device shown in FIG. 8 in another view.

[0016] FIG. 12 is a structural diagram of the antenna array device shown in FIG. 8 connected to a support.

DETAILED DESCRIPTION

[0017] It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. Additionally, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein can be practiced without these specific details. In other instances, methods, procedures and components have not been described in detail so as not to obscure the related relevant feature being described. The drawings are not necessarily to scale and the proportions of certain parts may be exaggerated to better illustrate details and features. The description is not to be considered as limiting the scope of the embodiments described herein.

[0018] Several definitions that apply throughout this disclosure will now be presented.

[0019] The term “coupled” is defined as connected, whether directly or indirectly through intervening components, and is not necessarily limited to physical connections. The connection can be such that the objects are permanently connected or releasably connected. The term “substantially” is defined to be essentially conforming to the particular dimension, shape, or another word that “substantially” modifies, such that the component need not be exact. For example, “substantially cylindrical” means that the object resembles a cylinder, but can have one or more deviations from a true cylinder. The term “comprising” means “including, but not necessarily limited to”; it specifically indicates open-ended inclusion or membership in a so-described combination, group, series, and the like.

[0020] An antenna system composed of many identical single antennas arranged in a certain pattern is also called an antenna array. A phased array antenna formed by the antenna array can efficiently communicate with low-orbit satellites due to its high flexibility and wide-angle scanning advantages. With the development of low-orbit satellite communication technology, wireless communication devices equipped with antenna arrays that can communicate with low-orbit satellites have also emerged. However, the antenna array has poor heat dissipation performance, especially in the narrow space of the wireless communication device. If the heat generated by the antenna array during operation cannot be quickly dissipated, the signal transmission quality will be affected.

[0021] Based on this, the present application provides a heat dissipation module and an antenna array device having the heat dissipation module, which can accelerate a heat dissipation speed of the antenna array.

[0022] Referring to FIG. 1, FIG. 1 illustrates a structural diagram of an antenna array device **10** according to an embodiment of the present application. The antenna array device **10** is used to achieve communication with one or more of low-orbit satellites, communication base stations, and wireless communication devices.

[0023] Referring to FIG. 2, the antenna array device **10** includes an upper cover **110**, a radiation module **120**, a circuit board **130**, a heat dissipation module **140**, and a lower cover **150**. The radiation module **120** is configured to transmit or receive signals to realize communications between the antenna array device **10** and other communication devices, such as low-orbit satellites, communication base stations, or wireless communication devices, etc. Furthermore, the radiation module **120** includes at least one radiation layer and at least one substrate layer. The radiation layer is provided with a plurality of radiation units. The plurality of radiation units cooperatively form the antenna array for effectively realize communications between the antenna array device **10** and the low-orbit satellites.

[0024] The antenna array includes a receiving antenna array and/or a transmitting antenna array. When the antenna array includes the receiving antenna array and the transmitting antenna array, a

part of the plurality of radiation units may form the receiving antenna array, and the other part of the plurality of radiation units may form the transmitting antenna array. The circuit board **130** is provided with radio-frequency circuits, control circuits, etc. for realizing communication control to the antenna array device **10**. The heat dissipation module **140** is in contact with the circuit board **130** for dissipating heat from the circuit board **130** and/or components on the circuit board **130** to decrease a temperature of the antenna array device **10** and ensure a normal operation of the antenna array device **10**. Referring to FIG. **3** together, the upper cover **110** and the lower cover **150** are connected and cooperatively form a receiving space **160**. The radiation module **120**, the circuit board **130**, and the heat dissipation module **140** are correspondingly overlapped and received in the receiving space **160**.

[0025] Referring to FIG. **4**, the heat dissipation module **140** includes a plurality of heat-conducting structures **141**, a plurality of supporting structures **142**, a plurality of connecting structures **143**, and a body **146**. The body **146** includes a first side and a second side that are opposite to each other. Please refer to FIG. **2** and FIG. **4**, a side of the body **146** close to the circuit board **130** is the first side, and a side of the body **146** away from the circuit board **130** is the second side. The plurality of heat-conducting structures **141** and the plurality of supporting structures **142** are disposed on the first side of the body **146**. The plurality of connecting structures **143** are disposed on an edge of the body **146**.

[0026] The plurality of heat-conducting structures **141** are arranged in an array on one side of the body **146**. Each heat-conducting structure **141** is made of a heat-conducting material. The heat-conducting material includes but is not limited to any one or a combination of two or more of heat-conducting plastics, ceramics, and metals. In some embodiments, the heat-conducting structures **141** may be made of one or two or more alloys of gold, silver, copper, iron, aluminum and tin. In other embodiments, the heat-conducting structures **141** may be made of one or a combination of any of ceramic materials such as silicon oxide, aluminum oxide, and zinc oxide.

[0027] Referring to FIG. **2**, in the embodiments of the present application, the radiation module **120** is disposed on a side of the circuit board **130** away from the heat dissipation module **140**, that is, the plurality of radiation units are disposed on a side of the circuit board **130** away from the heat dissipation module **140**. The RF circuit on the circuit board **130** may include a plurality of RF modules (not shown). The plurality of RF modules on the circuit boards **130** may be arranged one-to-one corresponding to the plurality of radiating units in the radiation module **120**. The RF module includes but is not limited to at least one of a waveform generator, a signal generator, an analog-to-digital conversion (ADC), a phase shifter, and a power divider. For instance, when the radiation module **120** includes 1024 radiation units, the RF circuit on the circuit board **130** may include corresponding 1024 RF modules. The radiation units form a first array, the RF modules form a second array, the first array and the second array correspond to each other. In this way, it is urgent to dissipate the heat from the RF modules on the circuit board **130** to maintain the normal operation of the RF modules and the corresponding radiation units. In the present application, the plurality of heat-conducting structures **141** are arranged on the other side of the circuit board **130** away from the plurality of radiation units. The plurality of heat-conducting structures **141** are arranged in an array, for example, forming a third array. The third array corresponds to the second array. Since the second array corresponds to the first array, the third array corresponds to the first array. In other words, the plurality of heat-conducting structures **141** are arranged corresponding to the plurality of radiation units. In this way, each heat-conducting structure **141** can respectively contact the circuit board **130** and/or the corresponding RF module on the circuit board **130**, and each heat-conducting structure **141** is used to conduct heat from the circuit board **130** and/or the corresponding RF module to itself, thereby reducing the temperature of the circuit board **130** and/or the RF modules, thereby ensuring the normal operation of the antenna array device **10**. In the present application, the RF modules are disposed on the side of the circuit board **130** close to the heat dissipation module **140**. In this way, the heat-conducting structures **141** can directly contact the RF modules to

conduct the heat generated by the RF modules to the outside.

[0028] In some embodiments, each heat-conducting structure **141** is a strip structure, and the plurality of heat-conducting structures **141** are arranged in rows to form an array. Thus, when the plurality of RF modules on the circuit board **130** are arranged in rows, the plurality of heat-conducting structure **141** can respectively correspond to the plurality of RF modules arranged in rows and simultaneously conduct heat. The present application does not limit the number of heat-conducting structures **141**. For example, in other embodiments, the plurality of heat-conducting structures **141** may also be arranged in a one-to-one correspondence with the plurality of RF modules, so that the number of the plurality of heat-conducting structures **141** may be equal to the number of the plurality of RF modules. In other embodiments, each heat-conducting structure **141** may also be a circular structure or a polygonal structure, etc., the present application does not impose any structural limitation on the heat-conducting structure **141**.

[0029] Referring to FIG. 5, FIG. 5 is a schematic diagram of the areas of the plurality of heat-conducting structures **141** and the areas of the plurality of radiation units, wherein S1 is the area of a heat-conducting structure among the plurality of heat-conducting structures **141**, and S2 is the area occupied by the plurality of radiation units. In some embodiments, a ratio of the area of the plurality of heat-conducting structures **141** (i.e., all S1) compared to the area of the plurality of radiation units (i.e., S2) is greater than or equals to 20%. In this way, the plurality of heat-conducting structures **141** can effectively dissipate heat for the antenna array device **10**, ensuring that the antenna array device **10** operates normally.

[0030] Referring to FIG. 4 again, in some embodiments, the heat dissipation module **140** further includes a plurality of heat conductive sheets **144**. The heat conductive sheets **144** are disposed between the RF modules and the heat conductive structures **141**, or the heat conductive sheets **144** are disposed between the circuit board **130** and the heat conductive structures **141**. A heat conductivity of the heat conductive sheet **144** is greater than or equal to 0.8 W/Mk. A hardness of the heat conductive sheet **144** is greater than or equal to 10 Shore, and the hardness of the heat conductive sheet **144** is less than or equal to 70 Shore. In other words, the heat conductive sheet **144** has a heat conductive property and has a certain elasticity. For example, the heat conductive sheet **144** includes but is not limited to any one of a heat conductive silicone sheet, a heat conductive tape, a heat conductive paste, a heat conductive putty, a heat conductive sealing glue, a heat conductive glass fiber cloth, a ceramic heat sink, a graphite sheet, graphene, a phase change material, and a composite material. When the heat conductive structure **141** is formed of a heat conductive material with greater hardness and the heat conductive structure **141** is in direct contact with the circuit board **130** or the RF modules on the circuit board **130**, the heat conductive structure **141** may damage the circuit board **130** or the RF modules. Therefore, in the embodiments, by providing the heat conductive sheets **144** between the heat conductive structures **141** and the RF modules (or the circuit board **130**), the heat of the circuit board **130** or the RF modules on the circuit board **130** can be transferred to the heat conductive structures **141** through the heat conductive sheets **144**, while reducing the probability of damage to the RF modules (or the circuit board **130**).

[0031] Accordingly, in some embodiments, at least one end of the heat-conducting structure **141** is further provided with a limiting portion **1411**. The limiting portion **1411** is used to limit the heat conductive sheet **144**. In one embodiment of the present application, a groove is provided on an upper surface of the heat conductive structure **141** close to the circuit board **130** to form the limiting portion **1411**. The heat conductive sheet **144** is disposed in the limiting portion **1411**. When the heat conductive sheet **144** is disposed in the limiting portion **1411**, a height of the surface of the heat conductive sheet **144** is greater than a height of the upper surface of the heat conductive structure **141**, and a thickness of the heat conductive sheet **144** is greater than a gap between the limiting portion **1411** and the RF module (or circuit board **130**). That is to say, two sides of the heat conductive sheet **144** are in close contact with an inner surface of the limiting portion **1411** and the

RF module (or the circuit board **130**) respectively. In this way, the heat conductive sheet **144** can transfer the heat on the RF module (or circuit board **130**) to the heat conductive structure **141**, and the heat conductive sheet **144** prevents the heat conductive structure **141** from directly contacting the RF module (or circuit board **130**), thereby reducing the probability of damage to the RF module (or circuit board **130**). In the embodiment, a length of the limiting portion **1411** is substantially the same as a length of the heat conductive structure **141**. Thus, only one groove needs to be formed on the surface of each heat conducting structure **141** to form the corresponding limiting portion **1411**. In other embodiments, grooves may be defined at both ends of the heat conductive structure **141** to form the limiting portions **1411**, or in other embodiments, three or more grooves may be defined on the heat conducting structure **141** to form the limiting portions **1411**. That is to say, the present application does not limit the quantity and location of the limiting portions **1411** on the heat conductive structures **141**. Accordingly, the quantity of heat conductive sheets **144** in the present application can be adjusted according to the quantity of limiting portions **1411**, and the present application does not limit the quantity of the heat conductive sheets **144**.

[0032] Referring to FIG. 4, each supporting structure **142** of the heat dissipation module **140** is used to support the circuit board **130**. In order to ensure the accuracy of phase control of the antenna array formed by the radiation module **120**, the radiation layer and the dielectric layer of the radiation module **120** should be prevented from shifting as much as possible. In the present application, the circuit board **130** is supported by the supporting structures **142** to fix the radiation module **120** between the circuit board **130** and the upper cover **110** (see FIG. 2), thereby improving the stability of the radiation module **120** and avoiding misalignment of the radiation layer or the dielectric layer of the radiation module **120**, which would affect the phase control accuracy and radiation efficiency of the antenna array device **10**. Specifically, the supporting structures **142** contact positions of the circuit board **130** where no radio frequency module or circuit is arranged. Thus, the probability of the supporting structures **142** damaging the circuit or electronic components on the circuit board **130** can be reduced. In some embodiments, the supporting structures **142** are made of metal material, so that the supporting structures **142** contact the positions of the circuit board **130** where no RF module or circuit is arranged, on the one hand, it can conduct heat to the circuit board **130**, and on the other hand, it can reduce the interference of the supporting structures **142** to the radiation module **120** and the circuit on the circuit board **130**.

[0033] Referring to FIG. 4, in some embodiments, each supporting structure **142** includes a first supporting portion **1421** and a second supporting portion **1422**. A diameter of the first supporting portion **1421** is greater than a diameter of the second supporting portion **1422**. One end of the first supporting portion **1421** is connected to the body **146**, the other end of the first supporting portion **1421** is connected to one end of the second supporting portion **1422**, and the other end of the second supporting portion **1422** abuts against the circuit board **130**. Since the supporting structures **142** are used to support the circuit board **130**, the supporting structures **142** need to reach a certain length to contact the circuit board **130** to play a supporting role. Further, in order to avoid the supporting structures **142** being too long and affecting the strength of the supporting structures **142**, a diameter of the supporting structure **142** is increased to increase the strength of the support structure **142**. In order to avoid a contact area between the supporting structures **142** and the circuit board **130** being too large and affecting the circuit layout design of the circuit board **130**, a structure with a small head and a large bottom is designed to form the first supporting portion **1421** and the second supporting portion **1422**. In this way, the support strength of the supporting structures **142** can be enhanced while reducing the impact of the supporting structures **142** to the circuit board **130**.

[0034] The first supporting portion **1421** and the second supporting portion **1422** may be integrally formed or separately connected, and are not limited by this application.

[0035] In other embodiments, the supporting structure **142** may also be a cylinder or other polygonal cylinders, etc. The present application does not limit the specific shape of the supporting

structure **142**.

[0036] In some embodiments, the antenna array device **10** further includes a plurality of buffers **145**. The buffers **145** are disposed between the supporting structures **142** and the circuit board **130**. The buffers **145** may be formed by elastic material. The buffers **145** are used to buffer a moment when the supporting structures **142** and the circuit board **130** (see FIG. 2) collide with each other, thereby reducing a probability of damaging the circuit board **130** due to an abutment between the supporting structures **142** and the circuit board **130**. That is, one end of the second supporting portion **1422** away from the first supporting portion **1421** may abut against the circuit board **130** through the buffers **145**. In some embodiments, the buffers **145** may be any one of sponge, rubber and foam.

[0037] The plurality of connecting structures **143** are used to connect the body **146** and the upper cover **110**. Through the connection between the connecting structures **143** and the upper cover **110**, the radiation module **120** and the circuit board **130** can be further fixed between the upper cover **110** and the heat dissipation module **140** (see FIG. 2). In this way, on the one hand, the heat dissipation module **140** can be in close contact with the circuit board **130**, thereby effectively conducting the heat of the circuit board **130** and the RF module on the circuit board **130**; on the other hand, the radiation module **120** and the circuit board **130** can be limited between the upper cover **110** and the heat dissipation module **140**, reducing the probability of displacement of the radiation layer and the dielectric layer of the radiation module **120**, thereby increasing the working accuracy of the antenna array device **10**. In some embodiments, the connecting structures **143** may be connecting holes formed on an edge of the body **146**. Accordingly, an edge of the inner side (i.e., a side close to the radiation module **120**) of the upper cover **110** is also provided with corresponding matching connection structures (not shown in the figures). For example, in some embodiments, connection holes are provided on the edge of the body **146** as the connecting structures **143**. The edge of the inner side of the upper cover **110** is provided with matching connection holes as the matching connecting structures. The connection between the body **146** and the upper cover **110** can be achieved by passing connecting members through the connecting structures **143** and the matching connecting structures. The connecting structures **143** can be made of metal material or non-metal material, which is not limited in the present application.

[0038] Please refer to FIG. 2 again, in some embodiments, a first side of the body **146** is further provided with a plurality of positioning posts **147**. The plurality of positioning posts **147** are used to achieve preliminary positioning between the heat dissipation module **140**, the circuit board **130**, and the upper cover **110**. Specifically, corresponding to the plurality of positioning posts **147**, the circuit board **130** is provided with a plurality of first positioning holes **131**. Mounting posts (not shown in the figures) are provided on the inner side of the upper cover **110**. A through hole is provided in the positioning post **147**, and a receiving hole is provided in the mounting post. The through hole in the positioning post **147** is aligned with the corresponding first positioning hole **131** and the receiving hole in the mounting column, and the positioning member passes through the through hole and the first positioning hole **131** and is partially received in the receiving hole of the mounting post, thereby preliminarily achieving the preliminary positioning of the upper cover **110**, the circuit board **130**, and the heat dissipation module **140**. The positioning post **147** can be made of metal material or non-metal material, and this application is not limited to this.

[0039] Referring to FIG. 6, in some embodiments, the heat dissipation module **140** further includes a plurality of heat sinks **148**. The plurality of heat sinks **148** are disposed at intervals at one end of a second side of the body **146**. The heat sinks **148** are also made of heat conductive materials and are generally in the shape of sheets. A plane where the heat sinks **148** are located is generally vertically connected to a plane where the body **146** is located. Correspondingly, the lower cover **150** is further provided with heat dissipation slots **151** corresponding to the plurality of heat sinks **148** (see FIG. 2). The heat sinks **148** are exposed from the lower cover **150** through the heat dissipation slots **151**. Thus, the heat-conducting structures **141** and the supporting structures **142** conduct heat to the heat

sinks **148** through the body **146**, so that the heat is transferred to the air outside the lower cover **150** through the heat sinks **148**, thereby effectively dissipating heat. In some embodiments, the plurality of heat sinks **148** also extend radially outward. The present application increases a surface area by providing the plurality of heat sinks **148**, thereby improving the efficiency of heat conduction to the air and improving the heat dissipation effect.

[0040] Please refer to FIG. 2 and FIG. 6, the heat dissipation module **140** further includes a plurality of bosses **149**. The plurality of bosses **149** surround the plurality of heat sinks **148** and are disposed on the second side of the body **146**. The plurality of bosses **149** are used to connect with the lower cover **150**. In one embodiment, each boss **149** is provided with a locking hole.

Accordingly, the lower cover **150** is provided with through holes around the heat dissipation slots **151**. The body **146** of the heat dissipation module **140** and the lower cover **150** can be connected by the locking piece passing through the through hole and being partially received in the locking hole.

[0041] Please refer to FIGS. 4 and 6, in some embodiments, the plurality of heat-conducting structures **141**, the plurality of supporting structures **142**, the plurality of connecting structures **143**, the body **146**, the plurality of positioning posts **147**, the plurality of heat sinks **148**, and the plurality of bosses **149** are all formed of heat-conducting materials, and the plurality of heat-conducting structures **141**, the plurality of supporting structures **142**, the plurality of connecting structures **143**, the body **146**, the plurality of positioning posts **147**, the plurality of heat sinks **148**, and the plurality of bosses **149** can be integrally formed by a CNC (Computer numerical control machine tools) process. In other embodiments, the body **146** is also made of the heat-conducting material. The plurality of heat-conducting structures **141**, the plurality of supporting structures **142**, the plurality of connecting structures **143**, the body **146**, and the heat sink **148** can be spliced to form the heat dissipation module **140**. In some embodiments, strip-shaped protrusions can be formed by bending the surface of the body **146** to serve as the heat-conducting structures **141**, and correspondingly, grooves **1461** are formed on the second side of the body **146** corresponding to the positions of the heat-conducting structures **141**, which is conducive to reducing a weight of the antenna array device **10**.

[0042] Please refer to FIG. 2 again, in some embodiments, the edge of the circuit board **130** is further provided with a plurality of second positioning holes **132**. Correspondingly, each dielectric layer and each radiation layer in the radiation module **120** is provided with a plurality of third positioning holes **121**. By passing a plurality of fasteners through the corresponding second positioning holes **132** and third positioning holes **121**, the positioning and connection between the circuit board **130** and the radiation module **120** can be achieved.

[0043] The lower cover **150** is also provided with a plurality of first mounting holes **152** on one side close to the heat dissipation module **140**. The upper cover **110** is also provided with second mounting holes (not shown) corresponding to the plurality of first mounting holes **152**. A plurality of mounting members pass through the corresponding first mounting holes **152** and are partially received in the second mounting holes, thereby achieving connection between the upper cover **110** and the lower cover **150**.

[0044] In some embodiments, the antenna array device **10** further includes a sub-circuit board **170**. The sub-circuit board **170** is disposed on a side of the heat dissipation module **140** away from the circuit board **130**. That is, the sub-circuit board **170** is disposed between the heat dissipation module **140** and the lower cover **150**. The sub-circuit board **170** is provided with a processor, a power conversion circuit, a modem board, a GPS (Global Positioning System) communication module, an interface, etc. The sub-circuit board **170** is electrically connected to the circuit board **130** for supplying power to the circuit board **130**. In some embodiments, the power conversion circuit can also be disposed on the circuit board **130**. The present application can reduce the area of the circuit board **130** by disposing the sub-circuit board **170**.

[0045] An installation process of the antenna array device **10** is roughly as follows:

[0046] First, please refer to FIG. 2 again, a plurality of fasteners connect the plurality of second

positioning holes **132** and the corresponding plurality of third positioning holes **121**, so as to realize the positioning and connection between the circuit board **130** and the radiation module **120**. Then, the through holes in the positioning posts **147** are aligned with the corresponding first positioning holes **131** and the receiving holes in the mounting posts, and the positioning members pass through the through holes and the first positioning holes **131**, and are partially received in the receiving holes of the mounting posts, so as to realize the preliminary positioning of the upper cover **110**, the circuit board **130**, and the heat dissipation module **140**. Then, referring to FIG. 7, the connecting members pass through the connecting structures **143** and the matching connecting structures on the inner side of the upper cover **110** to achieve the locking connection between the body **146** and the upper cover **110**. Next, the locking members pass through the through holes of the lower cover **150** and are partially received in the locking holes of the bosses **149** to achieve the preliminary connection between the body **146** of the heat dissipation module **140** and the lower cover **150**.

[0047] The first positioning holes **131**, the second positioning holes **132**, the third positioning holes **121**, the through holes, the receiving holes, the mounting holes, the locking holes, the first mounting holes **152**, and the second mounting holes mentioned in the present application may be threaded holes. Accordingly, the fasteners, the positioning members, the connecting members, the locking members, and the mounting members may be bolts, hot melt bolts, etc. The present application does not limit the connection or positioning method between the modules in the antenna array device **10**. In other embodiments, the connection or positioning can also be carried out by means of adhesives and/or snap-fitting.

[0048] Please refer to FIG. 7, in some embodiments, when the connecting structures **143** (or the positioning posts **147**) is made of metal materials, a height of the connecting structure **143** (or the positioning post **147**) is less than or equal to a height of the circuit board **130**. In this way, the interference of the connecting structures **143** on the radiation module **120** can be reduced.

Accordingly, in the present application, the upper cover **110** and the lower cover **150** are also made of insulating material to reduce the interference with the radiation module **120**.

[0049] The present application does not limit the specific shape of the body **146**. In other embodiments, the body **146** may also be a circular sheet or a polygonal sheet, etc.

[0050] In the present application, a waterproof gasket is further provided corresponding to the heat dissipation slots **151** and the connection between the upper cover **110** and the lower cover **150** to play a role of waterproofing, dustproofing, and reducing vibration.

[0051] In summary, the heat dissipation module **140** provided in the present application increases the heat conductivity of the heat-conducting structures **141** by setting the plurality of heat-conducting structures **141** to form an array, thereby effectively dissipating heat for the antenna array device **10**; at the same time, the circuit board **130** is supported by the plurality of supporting structures **142** and the plurality of connecting structures **143** are connected to the upper cover **110**, thereby reducing the probability of displacement of the radiation module **120** in the antenna array device **10** and improving the phase control accuracy of the antenna array device **10**.

[0052] Please refer to FIG. 8 and FIG. 9, another embodiment of the present application further provides an antenna array device **10a**. The antenna array device **10a** includes the upper cover **110**, the radiation module **120**, the circuit board **130**, a heat dissipation module **140a**, and a lower cover **150a**. A structure of the antenna array device **10a** is substantially the same as that of the antenna array device **10**, except that the heat dissipation module **140a** and the lower cover **150a** in the antenna array device **10a** are different from the structures of the heat dissipation module **140** and the lower cover **150** in the antenna array device **10**, and the placement position of the sub-circuit board **170** is different.

[0053] The structure of the heat dissipation module **140a** is substantially the same as that of the heat dissipation module **140** in the antenna array device **10**. The heat dissipation module **140a** also includes the plurality of heat-conducting structures **141**, the plurality of supporting structures **142**, the plurality of connecting structures **143**, the plurality of heat conductive sheets **144**, the plurality

of buffers **145**, the body **146**, and the plurality of positioning posts **147** (see FIG. 4). The difference between the heat dissipation module **140a** and the heat dissipation module **140** lies in the different structural arrangement of the second side of the body **146**, and in the heat dissipation module **140a**, the second side of the body **146** and the lower cover **150a** simultaneously serve as the bottom cover of the antenna array device **10a** (see FIG. 8).

[0054] Specifically, referring to FIG. 10, in this embodiment, the heat dissipation module **140a** includes a plurality of first heat sinks **1481a** and a plurality of second heat sinks **1482a**. The second side of the body **146** includes a first area **1462** and a second area **1463**. The first area **1462** is approximately located at a center of the second side. The body **146** is further provided with a receiving groove **1464** on one side of the first area **1462**, and the receiving groove **1464** is used to receive the sub-circuit board **170**. The second area **1463** surrounds the first area **1462** and the receiving groove **1464**.

[0055] The plurality of first heat sinks **1481a** are arranged in rows at intervals in the first area **1462**. A first channel **1465** is formed between every two adjacent rows. In each row, a second channel **1466** is formed between every two adjacent first heat sinks **1481a**. The first channel **1465** extends in a first direction (e.g., the Y direction in FIG. 10), and the second channel **1466** extends in a second direction (e.g., the X direction in FIG. 10). In this embodiment, the second direction is substantially perpendicular to the first direction.

[0056] The plurality of second heat sinks **1482a** are arranged at intervals in the second area **1463**, and the plurality of second heat sinks **1482a** are centered on the first area **1462** and extend outward in a generally radial manner. Each second heat sink **1482a** is generally arc-shaped, and the closer to the periphery of the body **146**, the greater the curvature of the edge of the second heat sink **1482a**. That is, each second heat sink **1482a** is substantially fin-shaped and disposed on the second side of the body **146**. Further, the plurality of second heat sinks **1482a** located on the side of the first area **1462** away from the receiving groove **1464** are disposed one-to-one with the adjacent plurality of first heat sinks **1481a**, so that the second channel **1466** continues to extend outward substantially radially in the second area **1463**. At least two of the second heat sinks **1482a** are disposed corresponding to the first channels **1465**, such that the first channels **1465** continue to extend within the second area **1463**.

[0057] In other embodiments, the plurality of second heat sinks **1482a** may also extend in the second area **1463** in a corrugated shape or other shapes, and the present application is not limited to this.

[0058] In other embodiments, only the plurality of first heat sinks **1481a** may be disposed at intervals on the second side of the body **146**. The plurality of first heat sinks **1481a** may extend outward in a radial, corrugated, or other shape.

[0059] In the heat dissipation module **140a**, the heat-conducting structures **141**, the supporting structures **142**, the connecting structures **143**, the body **146**, the positioning posts **147**, the first heat sinks **1481a**, and the second heat sinks **1482a** can be integrally formed from metal materials by a CNC (Computer numerical control machine tools) process, thereby improving the heat dissipation effect of the heat dissipation module **140a**.

[0060] The lower cover **150a** is used to cover the receiving groove **1464** to protect the sub-circuit board **170** in the receiving groove **1464**. In this embodiment, in order to accelerate the heat dissipation speed of the sub-circuit board **170**, a side of the lower cover **150a** close to the sub-circuit board **170** may also be provided with the plurality of heat-conducting structures, the plurality of supporting structures, and the plurality of heat conductive sheets (not shown in FIG. 10). A plurality of third heat sinks **1501a** are also disposed on a side of the lower cover **150a** away from the sub-circuit board **170** corresponding to the plurality of second heat sinks **1482a** in the second area **1463**. Thus, when the lower cover **150a** is covered on the receiving groove **1464**, the second heat sinks **1482a** located on the side of the first area **1462** close to the receiving groove **1464** are connected with the third heat sinks **1501a**, so that the second channels **1466** continue to

extend outward in a generally radial shape in the second area **1463** (see FIG. **11**).

[0061] In this way, the heat dissipation module **140a** provided in this embodiment is provided with the plurality of first heat sinks **1481a** and the plurality of second heat sinks **1482a**, the first channel **1465** and the second channel **1466** are formed on the second side of the body **146** through the plurality of first heat sinks **1481a** and the plurality of second heat sinks **1482a**, thereby further increasing the surface area of the heat dissipation module **140a** in contact with the air, thereby improving the heat dissipation efficiency of the heat dissipation module **140a**.

[0062] Please refer to FIG. **12**, in one embodiment, the antenna array device **10a** further includes a bracket **180**. The bracket **180** is used to adjust the height and angle of the antenna array device **10a**. The bracket **180** includes a base **181**, a support member **182**, an angle adjustment connector **183**, and a coupling member **184**. One end of the support member **182** is connected to the base **181**, and the other end is connected to the angle adjustment connector **183**. The angle adjustment connector **183** is connected to the second side of the body **146** through the coupling member **184**.

[0063] Specifically, the coupling member **184** includes a first coupling portion **1841**, a second coupling portion **1842**, and a third coupling portion **1843**. The first coupling portion **1841** is generally sheet-shaped. Both ends of the first coupling portion **1841** extend upward and then bend inward to form the second coupling portion **1842** and the third coupling portion **1843** respectively. Thus, the ends of the first coupling portion **1841** and the second coupling portion **1842** are at a height difference from the plane where the first coupling portion **1841** is located. In this embodiment, the second coupling portion **1842** and the third coupling portion **1843** are used to connect with the body **146**. The first coupling portion **1841** is used to connect with the angle adjustment connector **183**. In this way, there is a gap between the first coupling portion **1841** and the body **146**, so as to reduce the influence of the coupling member **184** on the heat dissipation effect of the heat dissipation module **140a**.

[0064] Furthermore, the base **181** is used to provide stable support for the antenna array device **10**. The support member **182** is a retractable member, and a height of the antenna array device **10a** is adjusted by adjusting a height of the support member **182**. The angle adjustment connector **183** is used to adjust an angle of the antenna array device **10a**. In this embodiment, the angle adjustment connector **183** can achieve angle adjustment through the cooperation between a fixed shaft and a rotating shaft (not shown in the figures). The angle adjustment connector **183** is not limited to the specific structure shown in FIG. **12**, and the present application does not limit the specific structure of the angle adjustment connector **183**. In other embodiments, other connectors for achieving angle adjustment may also be used, such as a movable hinge.

[0065] Thus, in this embodiment, by providing the bracket **180** on the antenna array device **10a**, the antenna array device **10a** can be adjusted in height and angle according to the use environment, thereby improving a flexibility of the antenna array device **10a** when in use.

[0066] The antenna array device **10** may also be provided with the bracket **180**, which will not be described in detail herein.

[0067] The embodiments shown and described above are only examples. Even though numerous characteristics and advantages of the present technology have been set forth in the foregoing description, together with details of the structure and function of the present disclosure, the disclosure is illustrative only, and changes may be made in the detail, including in matters of shape, size and arrangement of the parts within the principles of the present disclosure, up to and including the full extent established by the broad general meaning of the terms used in the claims.

Claims

1. A heat dissipation module applied in an antenna array device, the antenna array device comprising an upper cover and a circuit board, the heat dissipation module comprising: a body; a plurality of heat-conducting structures arranged in an array on one side of the body; a plurality of

- supporting structures supporting the circuit board, the plurality of heat-conducting structures and the plurality of supporting structures arranged on a same side of the body; and a plurality of connecting structures arranged on an edge of the body and connected to the upper cover.
- 2.** The heat dissipation module of claim 1, wherein the plurality of heat-conducting structures are made of heat-conducting materials.
- 3.** The heat dissipation module of claim 1, wherein each of the plurality of heat-conducting structures is a strip structure, the plurality of heat-conducting structures are arranged in rows to form the array.
- 4.** The heat dissipation module of claim 1, further comprising a plurality of heat conductive sheets, wherein at least one end of each of the plurality of heat-conducting structures is provided with a limiting portion, the limiting portion is configured to limit the plurality of heat conductive sheets.
- 5.** The heat dissipation module of claim 4, wherein a heat conductivity of each of the plurality of heat conductive sheets is greater than or equal to 0.8 W/Mk, a hardness of each of the plurality of the heat conductive sheet is greater than or equal to 10 Shore, and is less than or equal to 70 Shore.
- 6.** The heat dissipation module of claim 1, wherein the antenna array device comprises a plurality of radiation units arranged on a side of the circuit board, the plurality of heat-conducting structures are arranged on the other side of the circuit board away from the plurality of radiation units, the plurality of heat-conducting structures are corresponding to the plurality of radiation units, a ratio of an area of the plurality of heat-conducting structures compared to an area of the plurality of radiation units is greater than or equals to 20%.
- 7.** The heat dissipation module of claim 1, further comprising a plurality of heat sinks, wherein the plurality of heat sinks are arranged at intervals on another side of the body, the plurality of heat sinks extend radially outward.
- 8.** The heat dissipation module of claim 1, further comprising a plurality of buffers, wherein the plurality of buffers are arranged between the plurality of supporting structures and the circuit board, the plurality of buffers are configured to buffer a moment when the plurality of supporting structures and the circuit board collide with each other.
- 9.** The heat dissipation module of claim 1, further comprising a plurality of first heat sinks and a plurality of second heat sinks, wherein a side of the body that opposite to the plurality of heat-conducting structures and the plurality of supporting structures comprises a first area and a second area, the second area surrounds the first area, the plurality of first heat sinks are arranged in rows at intervals in the first area, plurality of second heat sinks are arranged at intervals in the second area, and the plurality of second heat sinks are centered on the first area and extend outward in a generally radial manner.
- 10.** An antenna array device comprising: an upper cover; a circuit board; and a heat dissipation module comprising: a body; a plurality of heat-conducting structures arranged in an array on one side of the body; a plurality of supporting structures supporting the circuit board, the plurality of heat-conducting structures and the plurality of supporting structures arranged on a same side of the body; and a plurality of connecting structures arranged on an edge of the body and connected to the upper cover.
- 11.** The antenna array device of claim 10, further comprising a lower cover and a plurality of radiation units wherein the plurality of radiation units are arranged on a side of the circuit board, the upper cover and the lower cover are connected and cooperatively form a receiving space, the plurality of radiation units, the circuit board, and the heat dissipation module are received in the receiving space.
- 12.** The antenna array device of claim 10, further comprising a sub-circuit board, wherein the sub-circuit board is disposed on a side of the heat dissipation module away from the circuit board, the sub-circuit board is electrically connected to the circuit board.
- 13.** The antenna array device of claim 10, wherein the plurality of heat-conducting structures are made of heat-conducting materials.

14. The antenna array device of claim 10, wherein each of the plurality of heat-conducting structures is a strip structure, the plurality of heat-conducting structures are arranged in rows to form the array.

15. The antenna array device of claim 10, wherein the heat dissipation module further comprises a plurality of heat conductive sheets, at least one end of each of the plurality of heat-conducting structures is provided with a limiting portion, the limiting portion is configured to limit the plurality of heat conductive sheets.

16. The antenna array device of claim 15, wherein a heat conductivity of each of the plurality of heat conductive sheets is greater than or equal to 0.8 W/Mk, a hardness of each of the plurality of the heat conductive sheet is greater than or equal to 10 Shore, and is less than or equal to 70 Shore.

17. The antenna array device of claim 11, the plurality of heat-conducting structures are arranged on the other side of the circuit board away from the plurality of radiation units, the plurality of heat-conducting structures are corresponding to the plurality of radiation units, a ratio of an area of the plurality of heat-conducting structures compared to an area of the plurality of radiation units is greater than or equals to 20%.

18. The antenna array device of claim 10, wherein the heat dissipation module further comprises a plurality of heat sinks, the plurality of heat sinks are arranged at intervals on another side of the body, the plurality of heat sinks extend radially outward.

19. The antenna array device of claim 10, wherein the heat dissipation module further comprises a plurality of buffers, the plurality of buffers are arranged between the plurality of supporting structures and the circuit board, the plurality of buffers are configured to buffer a moment when the plurality of supporting structures and the circuit board collide with each other.

20. The antenna array device of claim 10, wherein the heat dissipation module further comprises a plurality of first heat sinks and a plurality of second heat sinks, a side of the body that opposite to the plurality of heat-conducting structures and the plurality of supporting structures comprises a first area and a second area, the second area surrounds the first area, the plurality of first heat sinks are arranged in rows at intervals in the first area, plurality of second heat sinks are arranged at intervals in the second area, and the plurality of second heat sinks are centered on the first area and extend outward in a generally radial manner.
